

Chapter 2

THE INTEGUMENT

Reviews of various general aspects of the insect integument have been given by Beament (1961, 1964), Hackman (1971, 1974), Hinton (1970), Lawrence (1967), Locke (1964, 1967), Neville (1967, 1970, 1975), Weis-Fogh (1970) and Wigglesworth (1959).

Structure, Composition and Functions

The integument consists of the following layers: (i) the cuticle, (ii) the epidermis, and (iii) the basement membrane (Fig. 2).

(i) **The Cuticle** is a complex, non-cellular layer secreted largely by the epidermis and though commonly considered non-living is actually the seat of complex biochemical changes, some at least under enzymatic control. It forms the outermost investment of the body and its appendages but is invaginated locally to form endoskeletal structures (p. 81) and also provides

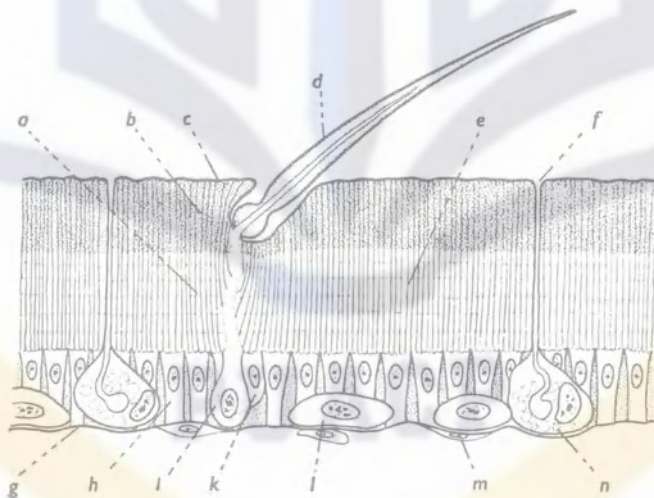


FIG. 2 Section of typical insect cuticle (after Wigglesworth)

a, laminated endocuticle; b exocuticle; c, epicuticle; d, bristle; e, pore-canals; f, duct of dermal gland; g, basement membrane; h, epidermal cell; i, trichogen cell; k, tormogen cell; l, oenocyte; m, haemocyte adherent to basement membrane; n, dermal gland. From Wigglesworth, *Principles of Insect Physiology*, 7th edn, p. 27, Fig. 19.

the lining of the tracheal system, some glands and parts of the alimentary canal and reproductive tract. When newly formed it is flexible and elastic and in many larvae it remains so over much of the body. In most insects, however, the greater part of the cuticle undergoes a process of sclerotization whereby it becomes hardened and darkened to form more or less tough, rigid sclerites separated from each other by membranous zones of unchanged soft cuticle. Such an arrangement combines rigidity with flexibility and in addition to its protective function the cuticle determines the form of the insect, its relative impermeability to water reduces desiccation and it provides a firm basis for the attachment of muscles. Two main layers may be discerned in the cuticle, an outer epicuticle and an inner procuticle, both generally compound structures. The epicuticle is a very thin layer devoid of chitin and of complex chemical composition and ultrastructure. In the terminology of Locke (1964, 1966) and Filshie (1970) there is a dense outer cuticulin layer about 10 nm thick which is very resistant to acids and organic solvents and probably contains a highly polymerized lipid. It is the first layer to be secreted in the cycle of cuticle formation, arising on top of the epidermal microvilli as small plaques which subsequently fuse to cover virtually the whole body. Beneath it lies the main epicuticular layer, the so-called protein epicuticle, also containing bound lipid. This is a homogeneous, dense, refractile layer about 1 μm thick, formed from granules that originate in the Golgi complex of the epidermal cells. It is the material of this layer to which the term 'cuticulin' was originally applied by Wigglesworth (1947). In addition the epicuticle may include a superficial lipid layer (Lockey, 1960) covered by a protective cement layer of unknown composition; the cement layer is secreted by epidermal glands and poured out over the surface. Before ecdysis the epicuticle is folded but its shape determines the overall shape of the insect (Bennet-Clark, 1963) and it bears a more or less elaborate surface pattern (Locke, 1967). Primitively this pattern comprises a network of polygons, each the product of a single epidermal cell, though such a simple arrangement is often replaced by other patterns of special functional significance (Hinton, 1970).

The procuticle, which may be absent from the tracheoles, is secreted by the epidermal cells and makes up the bulk of the integument. The mechanical properties of insect cuticle are well adapted to its skeletal function and are determined largely by the various procuticular components, mainly chitin and various proteins (Rudall and Kenchington, 1973). Together they show a high resistance to impact, high tensile, flexural and compressive strengths and a high strength-weight ratio, the latter associated with the lack of mineral salts. Typically the procuticle comprises an outer exocuticle, composed of a homogeneous electron-dense matrix, hardened through sclerotization (see below), and shed entirely when the insect moults. Within this is a softer endocuticle, capable of increasing considerably through postecdysial deposition, and showing an elaborate ultrastructure in which variously oriented protein and chitin fibres form repeated layers 0.1 to 1 μm thick.

Especially characteristic of some endocuticles is the progressive 'helical' change of direction of these fibres in successive lamellae, which produces parabolic figures in sectioned material (Neville, 1970). Deposition of the endocuticle often occurs in daily growth-layers (Neville, 1963). These are visible by phase-contrast and polarization microscopy and occur in many Pterygote insects; such layers are not found in Apterygotes and are laid down with an irregular rhythm in the Coleoptera. Two further procuticular layers have been recognized: a fuchsinophil mesocuticle (Schatz, 1952) lying between endo- and exocuticle, and a basal subcuticle (Schmidt, 1956) whose granular ultrastructure suggests that the newly secreted cuticular microfibrils have not undergone orientation.

The procuticle is usually pierced by very numerous pore-canal which run perpendicularly to its surface and are initially occupied by cytoplasmic filaments from the epidermis, though they may later become filled with cuticular material. They are less than $1\text{ }\mu\text{m}$ in diameter and have a flat or twisted ribbon-like shape, through which the axial filaments run a straight course (Neville, Thomas and Zelazny, 1969; Neville and Luke, 1969a). The procuticle is probably secreted around them and they may also transport material to the outer procuticle and epicuticle. Pore-canal are absent from the rubber-like cuticle found in some insects. Sclerotization, whereby the exocuticle acquires its hard, tough, inelastic character, is due to a process of tanning, in which adjacent protein chains are cross-linked by *o*-quinones. The events leading up to this have been much studied in the development of the Dipteran puparium, in which the quinones apparently arise by oxidation of a phenolic substrate *N*-acetyldopamine, itself derived from tyrosine (Karlson and Sekeris, 1966). Darkening of the hardened regions is a separate but related process; both are involved in the repair of cuticular injuries.

The two major components of insect cuticle are the carbohydrate chitin, which accounts for 25–60 per cent of the dry weight of various cuticles, and a number of proteins. On hydrolysis, chitin yields acetic acid and glucosamine and is a high molecular weight polymer consisting mainly of anhydro-*N*-acetylglucosamine residues joined by β -1,4-linkages, though up to 10 per cent of the residues may be deacetylated. The chains are unbranched and consist of several hundred units with a repeating structure $3\cdot1\text{ nm}$ long. There are three different crystallographic forms of chitin – α -, β - and γ -chitin – which differ in the arrangement of the chains and in the presence of bound water; α -chitin is the most stable and the only one found in arthropod cuticle. The metabolic pathways through which chitin is formed are not fully established but enzymatic synthesis of UDP-*N*-acetylglucosamine from glucose has been demonstrated by Candy and Kilby (1962) and there is evidence for polymerization under the control of a chitin synthetase. In the cuticle, chitin chains are apparently joined to proteins by covalent linkages involving aspartic acid and histidine (Rudall, 1963). The chitin-protein complex is in fact a polydisperse glycoprotein in which rod-like chitin fibres $2\cdot5$ to $6\cdot5\text{ nm}$ in diameter are embedded in a protein matrix (Neville and Luke, 1969b).

Chitin is insoluble in water, alkalis, dilute acids and organic solvents but dissolves with decomposition in concentrated mineral acids and sodium hypochlorite. It has a

specific gravity of about 1.4, a refractive index of about 1.55 and is best detected by the van Wisselingh test – treatment with concentrated potassium hydroxide at 160° C for 20 minutes converts it to chitosan which gives a rose-violet colour with 0.2 per cent iodine in 1 per cent sulphuric acid (Campbell, 1929). A more specific histological test depends on the use of conjugates of a chitinase with fluorescent dyestuffs (Benjaminson, 1969). The main structural proteins of the insect cuticle are sometimes known collectively as arthropodin though they are a complex mixture, more than a dozen components having been identified in some insects by amino-acid analyses, electrophoresis and serological techniques (Fox and Mills, 1969; Hackman and Goldberg, 1971). The proteins differ in amino-acid composition from one species

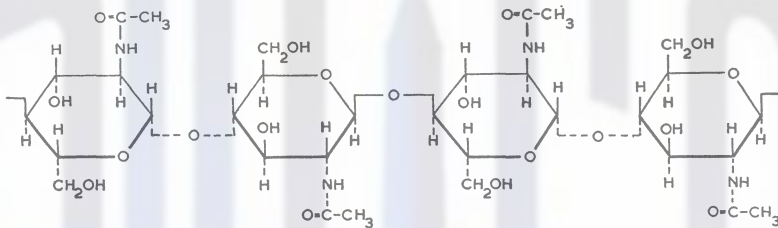


FIG. 3. Structural formula of chitin

to another but those of related species show some resemblances. Protein deposition in the cuticle is related to changes in the amino acids of the blood though the mechanisms of synthesis are not known. Of special interest is the distinctive protein resilin (Andersen and Weis-Fogh, 1964; Andersen, 1970). This resembles the elastin of vertebrate connective tissue, being an isotropic three-dimensional network of polypeptide chains held together by stable covalent cross-links and showing remarkable rubber-like properties. It is structureless and metabolically inert but forms mechanical 'springs' which are readily deformed and show perfect elastic recovery. It occurs in localized zones where these properties are functionally important, for example near the wing articulation of many insects and in the metathorax of fleas, where it forms part of the jumping mechanism (p. 945).

Other components of the cuticle account for only a very small part of its weight but are of great physiological significance. Phenolic precursors of the quinones that link amino-acid chains to form sclerotin are widely distributed, though some Apterygotes lack them and harden their cuticles by disulphide linkages (Krishnan, 1969). There are also various waxes and other lipids secreted by the oenocytes (Locke, 1969a) and responsible for waterproofing the cuticle (Beament, 1964; Gilby, 1965; Gilbert, 1967). These form a layer at or near the surface or are incorporated into inner layers of the cuticle. They are a complex mixture of up to 80 per cent hydrocarbons with smaller amounts of fatty acids, alkyl esters and other constituents. Chemically the waxes found in exuviae of *Bombyx mori* comprise a mixture of odd-numbered C₂₇–C₃₇ paraffins with esters of even-numbered C₂₆–C₃₀ fatty acids. Though the epicuticular lipids play a major role in reducing water loss through the integument, it now seems unlikely that this depends on an oriented layer of wax molecules subject to disruption at a critical transition temperature (Hackman, 1971). The integument also includes enzyme systems responsible for the synthesis and degradation of the major cuticular materials and those involved in

melanization and sclerotization; many of these systems require fuller investigation. Inorganic cuticular constituents are rare but calcareous nodules may develop on the outside of Stratiomyid and Psychodid larvae while *Rhagoletis cerasi* larvae show intra-cuticular calcification (Wiesmann, 1938). Pigments occurring in the cuticle are discussed below.

(2) **The Epidermis** forms a continuous single layer of cells, the plasma membranes of which are joined by numerous septate desmosomes. The ultrastructure of the cells varies with their cycles of secretory activity (Locke, 1969*b*) but they have a microvillate surface and contain numerous mitochondria, Golgi vesicles and cisternae of smooth-surfaced endoplasmic reticulum as well as cytoskeletal structures in the form of oriented microfibres and microtubules. Rough endoplasmic reticulum is present mainly in cells secreting resilin. Scattered among the normal epidermal cells are specialized gland cells and those concerned in the formation of cuticular sensilla (see p. 124). Muscle attachments penetrate the epidermis, the myofibrillae usually being associated with tonofibrillae that run through the epidermis and into the procuticle, while the oenocytes (p. 256) which originate from epidermal cells sometimes remain closely associated with this layer. Not only does the epidermis secrete the greater part of the cuticle but it also produces the moulting fluid (Bade and Wyatt, 1962; Jeuniaux, 1963), which dissolves the old endocuticle before the immature insect moults (p. 361), it absorbs the digestion products of the old cuticle, repairs wounds and differentiates in such a way as to determine the surface patterns of the insect (Wigglesworth, 1959; Lawrence, 1967).

(3) **The Basement Membrane** is a continuous layer about $0.5\ \mu\text{m}$ thick with an amorphous ultrastructure. It contains neutral mucopolysaccharides and is secreted by haemocytes (Wigglesworth, 1956). Chordotonal organs (p. 127), tracheoles, and nerves run to it or through it.

Cuticular Appendages

These structures include all outgrowths of the cuticle that are connected with it by means of a membranous joint. They arise from modified epidermal cells and may be classified into setae and spurs.

Setae or **Macrotrichia** (Fig. 4) are commonly known as hairs and each arises from a cup-like pit or *alveolus*. At its base the seta is attached by a ring of articular membrane. Setae are hollow structures developed as extensions of the exocuticle and each is produced by a single, usually enlarged, *trichogen cell*. The articular membrane is usually produced by a separate *tormogen cell*. The arrangement of the more constantly located setae (chaetotaxy) is important in the systematics of some insect groups, e.g. the Diplura, Thysanoptera, Cyclorrhaphan Diptera and larval Lepidoptera. The main kinds of setae and their modifications are listed below; the structure of some is now known in great detail through the use of specialized optical techniques and of the scanning and transmission electron microscopes (e.g. Baker and McCrae, 1966–67; Evans, 1967; Hale and Smith, 1966; Lippert and Gentil,

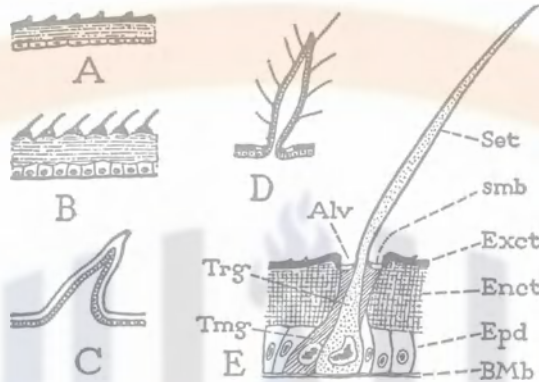


FIG. 4 External processes of the body-wall, diagrammatic. A, B, non-cellular cuticular processes. C, D, multi-cellular processes. E, a typical unicellular process or seta

Alv, setal socket or alveolus; *Set*, seta; *smb*, setal membrane; *Tmg*, tormogen or socket-forming cell; *Trg*, trichogen or seta-forming cell. Snodgrass, *Principles of Insect Morphology*, McGraw-Hill, 1935, Fig. 28.

1959; Lukoschus, 1962; Picken, 1949). (1) *Clothing hairs* – These invest the general surface of the body or its appendages and frequently exhibit various degrees of specialization. When furnished with thread-like branches as in the Apoidea they are termed *plumose hairs*. Setae which are particularly stout and rigid are known as *bristles* and are well exhibited for example in the Tachinidae. (2) *Scales* – These are highly modified clothing hairs and are characteristic of all Lepidoptera and many Collembola: they are also present in some Thysanura, Coleoptera, Diptera and Hymenoptera. Transitional forms between ordinary clothing hairs and scales are frequent. (3) *Glandular setae* – Grouped under this heading are those setae which function as the outlet for the secretion of epidermal glands (see p. 263). If they are especially stout and rigid they are then termed *glandular bristles* as in the urticating hairs of certain lepidopterous larvae. (4) *Sensory Setae* – Very frequently the setae of certain parts of the body, particularly the appendages, are modified in special ways and become sensory in function. Sensory setae (see p. 125) are in all cases connected with the nervous system.

Spurs occur on the legs of many insects and differ from setae in being of multicellular origin.

Cuticular Processes

The external surface of the cuticle, in addition to being sculptured in various ways, bears a great variety of outgrowths which are integral parts of its substance. They are rigidly connected with the cuticle, having no membranous articulation and are therefore readily separable from cuticular appendages. The principal types of cuticular processes are as follows.

Microtrichia (fixed hairs or aculei) – These are minute hair-like structures found, for example, on the wings of the Mecoptera and certain

Diptera. They resemble very small covering hairs, but the absence of the basal articulation is their distinguishing feature (Figs. 4 and 26).

Spines – This expression has been used by various writers with considerable latitude but is here confined to outgrowths of the cuticle which are more or less thorn-like in form. Spines differ from spine-like setae in being produced by undifferentiated epidermal cells and are usually, if not always, of multicellular origin.

In addition to the above there are many other cuticular processes which either take the form of more or less conical *nodules* and *tubercles* of different shapes, or of larger projections known as *horns* which are characteristic of the males of certain Coleoptera.

Coloration

The colours of adult and immature insects may be grouped into three classes: (1) pigmentary or chemical colours, (2) structural or physical colours, and (3) combination or chemico-physical colours.

1. Pigmentary Colours – These are due to chemical substances that absorb some wavelengths of the incident light and reflect others. They may be present in the cuticle, epidermis or subepidermal tissues (usually fat-body or blood). A colour pattern often consists of an epidermal or subepidermal ground colour (which may fade rapidly after death) and overlying areas of more permanent cuticular pigmentation. Eye pigments occurring in the ommatidial cells of some insects may be of special interest since studies of their metabolism reveal something of the mode of action of genes controlling eye colour.

Biochemical aspects of insect pigmentation have been reviewed recently by Cromartie (1959), Thomson (1962), Kilby (1963), Goodwin (1971), Fuzeau-Braesch (1972) and others. The substances concerned may be classified as follows.

(a) *Melanins*. These are amorphous, highly stable, dark brown or black cuticular pigments which are generally non-granular and are insoluble in the usual solvents though they are rapidly decolorized by oxidizing agents. Their chemical nature has not been satisfactorily elucidated and it is, in fact, difficult to distinguish between cuticular darkening due to sclerotization and that caused by the presence of true melanins (Richards, A. G., 1967). The mode of formation of melanins is uncertain (Lerner and Fitzpatrick, 1950) but they probably arise through polymerization of indole compounds which are derived initially from the amino-acid tyrosine by ring-closure and oxidation under the influence of the enzyme tyrosinase. Thus, *Aedes* larvae reared on a diet deficient in tyrosine or the related phenylalanine together with the resulting pupae are unpigmented (Golberg and De Meillon, 1948). Where melanic pigmentation is discontinuous (e.g. dark spots, etc.) this is apparently the result of a localized distribution of the substrate tyrosine, the tyrosinase occurring also in areas which remain unpigmented.

(b) *Carotenoids*. These are polyene pigments usually containing 40 carbon atoms in the molecule; they are readily soluble in fat-solvents and are characteristically

synthesized by plants (Lederer, 1938; Feltwell and Rothschild, 1974). When ingested by animals they accumulate in the blood and tissues unchanged or after minor oxidative alterations and in some cases they form the prosthetic group of a chromoprotein. β -carotene, derived from the tissues of the potato plant, occurs in the blood of the Colorado Potato-beetle *Leptinotarsa decemlineata* and is responsible for the red and yellow coloration of the Pentatomid *Perillus bioculatus* which preys on larvae of the beetle (Palmer and Knight, 1924). Again, in *Coccinella*, the red colour is due to the presence of the plant carotenoids lycopene and α - and β -carotene (Lederer, 1938). Astaxanthin (3:3-dihydroxy-4:4-diketo- β -carotene) and β -carotene both occur as chromoproteins in the integument of locusts and the green chromoprotein pigment of many insects (known as insectoverdin) is a complex, the yellow-orange component of which may have as its prosthetic group β -carotene (*Carausius*), lutein (*Sphinx*, *Tettigonia*) or astaxanthin.

(c) *Pteridines*. These pigments are derived from a heterocyclic pyrimidine-pyrazine ring structure and are found more often in insects than in other organisms (Ziegler and Harmsen, 1969). The majority belong to the class of pterines (2-amino-4-hydroxy-pteridines) and the natural pigment is often a mixture of two or more of these. Thus the conspicuous colours of the Heteroptera *Dysdercus*, *Pyrrhocoris*, *Oncopeltus* and *Phonoctonus* are due to the red and yellow pterines erythropterin and xanthopterin, along with the white leucopterin. Xanthopterin and leucopterin are also found in the integument of the Vespidae, while the body- and wing-scales of many Pierid butterflies contain leucopterin, xanthopterin, isoxanthopterin and erythropterin. Some pterines occur characteristically as eye pigments, e.g. a group of so-called drosopterins contribute to the red eye colour of the wild-type *Drosophila melanogaster*. Integumentary pterines lie in the epidermis and interlamellar spaces of scales and hairs as ellipsoidal granules about 0.5 μm long. In the ommatidia they occur as proteinaceous granules in the pigment cells surrounding the crystalline cone and retinulae; they are distinct from and sometimes masked by the ommochrome pigments that may also be present. In addition to their role as pigments some hydrogenated pterines can act metabolically as co-factors in hydroxylation reactions; in the Pieridae they are not only pigments but also represent the insoluble end-products that accumulate in 'storage excretion' of nitrogenous metabolites.

(d) *Ommochromes* (Linzen, 1974). These are best known as eye pigments and comprise two groups of substances, the sulphur-free ommatins of low molecular weight and the sulphur-containing ommins of high molecular weight. The latter are the main pigments occurring in the eye of insects from many orders. Xanthommatin is an eye pigment in *Drosophila*; its biosynthesis from tryptophane via kynurenine and 3-hydroxykynurenine is under genetic control. Like other ommochromes it is a redox pigment, yellow when oxidized and red when reduced. Integumentary ommochromes occur in the wings of Nymphalid butterflies and in *Schistocerca* and they are also involved in the colour changes of the stick insect *Carausius morosus*. In the dragonfly *Sympetrum striolatum* the scarlet male contains ommatins and the yellow female their reduced equivalents.

(e) *Anthraquinones*. These are confined to the Coccoidea, where the best known is carminic acid, the colouring principle of cochineal, found in the eggs and fat body of the female *Dactylopius coccus* and accounting for half its body weight. Kermesic acid is the pigment in the dyestuff kermes, from females of *Kermococcus ilicis*, and laccaic acid is the water-soluble red pigment of lac (p. 266).

(f) *Aphins*. These are polycyclic quinones which decompose immediately after

the death of the insect containing them, giving rise to the erythroaphin reported from *Aphis fabae*, *Eriosoma lanigerum* and other aphids.

(g) *Miscellaneous pigments*. Many other substances play a minor role as insect pigments, occurring either in small amounts or only in a few species. Among such are haemoglobin, derivatives of chlorophyll, anthoxanthins, anthocyanins, riboflavin (which accumulates as the greenish-yellow 'entomo-urochrome' of the Malpighian tubules) and purines. The bile pigment mesobiliverdin is perhaps more important since it is the prosthetic group of the blue component of the insectoverdins (complex green pigments) mentioned above in connection with the carotenoids.

2. Structural Colours – Structural colours differ from those due to pigments in that they are changed or destroyed by physical changes in the cuticle such as result from shrinkage, swelling, distortion or permeation with liquids of the same refractive index as the cuticle. They may also be duplicated by physical models, are not destroyed by bleaching and all the component wavelengths of the incident light are to be found in either the reflected, scattered or transmitted fractions. Of the many papers on this topic, see especially Mason (1926–27) and Anderson and Richards (1942). Four types of structural coloration may be distinguished: (i) *Structural white* is caused by the scattering, reflection and refraction of light by microscopic particles large in comparison with the wavelength of light and which, in themselves, are usually transparent. Probably most insect whites have a structural basis. (ii) *Tyndall blue*, though uncommon, occurs in some Odonata and is due to the scattering of the shorter wavelengths by particles with dimensions of about the same size as the wavelengths of light. (iii) *Interference colours* are produced by optical interference between reflections from a series of superimposed laminae or ribs. This is one of the commonest types of physical coloration and the iridescent appearance is well seen, for example, in the wing-scales of *Morpho* butterflies, in the Diamond beetles *Entimus* and *Cyphus* and in the Chrysidae. (iv) *Diffraction colours*, resulting from the presence of closely spaced striae, 0.5 to 3 μm apart, occur in the Mutillidae and various beetles such as *Serica*, several Carabidae and Gyrinidae, *Nicrophorus* and others (Hinton and Gibbs, 1969, 1971). It is possible that variations in the colour and intensity of light reflected from these structures help to confuse predators as to the size and distance of the insect.

The beetle *Dynastes hercules* can change colour quite quickly from yellow to black and vice versa. This process has a structural basis elucidated by Hinton and Jarman (1973). Below the transparent epicuticle of the elytra is a very thin spongy layer; when its interstices are filled with air this appears yellow, but when they fill with water the black colour of the underlying cuticle becomes apparent.

3. Combination Colours – These are produced by a structural modification in conjunction with a layer of pigment and are much commoner than purely structural colours. In the butterfly *Teracolus phlegyas* a red pigment in the scale wall (but not in the striae) combines with a structural violet to

produce magenta: in *Ornithoptera poseidon* the emerald green is due to a structural blue combined with a yellow pigment in the walls and striae of the scales. In a number of cases (e.g. Lycaenids) there is no indication of the cause of colour. The golden iridescence of *Cassida* and its allies is produced by a film of moisture beneath the surface cuticle. These insects rapidly lose their colour when dried, but it returns after soaking in water provided the drying has not been too prolonged.

Literature on the Body-wall and Coloration

- ANDERSEN, S. O. (1970), Resilin, In: Florkin, M. and Stotz, E. H. (eds), *Comprehensive Biochemistry*, 26C.
- ANDERSEN, S. O. AND WEIS-FOGH, T. (1964), Resilin. A rubberlike protein in Arthropod cuticle, *Adv. Insect Physiol.*, 2, 1-65.
- ANDERSON, T. F. AND RICHARDS, A. G. (1942), An electron-microscope study of some structural colours in insects, *J. appl. Phys.*, 13, 748-758.
- BADE, M. L. AND WYATT, G. R. (1962), Metabolic conversions during pupation of the cecropia silkworm. I. Deposition and utilization of nutrient reserves, *Biochem. J.*, 83, 470-478.
- BAKER, J. R. AND MCCRAE, J. M. (1966), The ultrastructure of the 'Podura' scale (*Lepidocyrtus curvicolis*, Collembola), as revealed by whole mounts, *J. Ultrastruct. Res.*, 15, 516-521.
- (1967), A further study of the *Lepidocyrtus* ('Podura') scale (Insecta: Collembola), *J. Ultrastruct. Res.*, 19, 611-615.
- BEAMER, J. W. L. (1961), The water relations of insect cuticle, *Biol. Rev.*, 36, 281-320.
- (1964), The active transport and passive movement of water in insects, *Adv. Insect Physiol.*, 2, 67-129.
- BENJAMINSON, M. A. (1969), Conjugates of chitinase with fluorescein isothiocyanate or lissamine rhodamine as specific stains for chitin *in situ*, *Stain Technol.*, 44, 27-31.
- BENNET-CLARK, H. C. (1963), The relation between epicuticular folding and the subsequent size of an insect, *J. Insect Physiol.*, 9, 43-46.
- CAMPBELL, F. L. (1929), The detection and estimation of insect chitin; and the interrelation of chitinization to hardness and pigmentation of the cuticle of the American cockroach, *Periplaneta americana* L., *Ann. ent. Soc. Am.*, 22, 401-426.
- CANDY, D. J. AND KILBY, B. A. (1962), Studies on chitin synthesis in the desert locust, *J. exp. Biol.*, 39, 129-140.
- CROMARTIE, R. I. (1959), Insect pigments, *A. Rev. Ent.*, 4, 59-76.
- EVANS, J. J. T. (1967), The integument of the Queensland fruit fly, *Dacus tryoni* (Frogg.), 1. The tergal glands, *Z. Zellforsch. mikrosk. Anat.*, 81, 49-61. 2. Development and ultrastructure of the abdominal integument and bristles, *Ibid.*, 81, 34-48.
- FELTWELL, J. AND ROTHSCHILD, M. (1974), Carotenoids in thirty-eight species of Lepidoptera. *J. Zool.*, 174, 441-465, 1 pl.
- FILSHIE, B. K. (1970), The fine structure and deposition of the larval cuticle of the sheep blowfly (*Lucilia cuprina*), *Tissue and Cell*, 2, 479-498.
- FOX, F. R. AND MILLS, R. R. (1969), Changes in haemolymph and cuticle proteins